Error-budgeted post-processing of randomized interpolative decompositions*

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Abstract

Randomized pivoting algorithms such as robust blockwise random pivoting (RBRP) compute interpolative decompositions (IDs) with small approximation error, but the skeleton they return can have a large interpolation matrix $W = XX_S^\dagger$. Its spectral norm, the *amplification* $A(S) = \|W\|_2$, governs the worst-case magnification of perturbations of the skeleton when the ID is used. We show that the approximation error does not control this quantity, and we study post-processing of an RBRP skeleton under an explicit error budget $E(S) \le (1+\varepsilon)E(S_0)$. We describe a deterministic exchange refinement that replaces one skeleton row at a time and accepts an exchange only if it strictly decreases the exact amplification within the budget; a scan budget T trades cost for benefit. We compare it with volume-based exchange (as in strong rank-revealing QR and maxvol), which is cheaper but does not control the error, and we propose a two-stage pipeline that runs a guarded volume exchange followed by the guarded refinement. On synthetic matrices, snapshot matrices and public real data (handwritten digits, image patches, an ECG delay matrix, a kernel matrix), the pipeline lowers the mean amplification of RBRP skeletons to between 29% and 79% of its initial value within a 5% error budget, whereas unguarded volume exchange violates the same budget in 32% of the runs. A full scan of the refinement gives a further reduction at a cost that grows roughly quadratically with the matrix size. When the skeleton rows are perturbed, the lower amplification reduces the reconstruction error once the perturbation exceeds a matrix-dependent level.

Key words. interpolative decomposition, randomized pivoting, skeleton selection, interpolation stability, strong rank-revealing QR, maximum volume

1 Introduction

Let $X \in \mathbb{R}^{n\times d}$ and let $S \subset \{1,\dots,n\}$ be a set of k *skeleton* rows, $X_S = X(S,:)$. A (row) interpolative decomposition (ID) approximates every row of X by a combination of the skeleton rows,

$$X \approx WX_S, \qquad W = XX_S^\dagger \in \mathbb{R}^{n\times k}, \tag{1}$$

where W contains an identity block in the rows of S [1]. The ID is the building block of CUR decompositions, skeletonization in fast direct solvers, and empirical interpolation in model reduction [15, 13].

Randomized pivoting. Choosing S by (strong) rank-revealing QR gives guarantees on W but is expensive [6]. Randomized pivoting algorithms select the skeleton adaptively from sketches or residuals and are fast [3, 4, 5]. RBRP is a blockwise randomized pivoting algorithm for adaptive

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interpolative decomposition [2]. The analysis and the experiments of that line of work concentrate on the approximation error

$$E(S) = \frac{\|X - WX_S\|_F}{\|X\|_F}. \tag{2}$$

Interpolation amplification. The interpolation itself is governed by

$$A(S) = \|W\|_2 = \|XX_S^\dagger\|_2 \;\geq\; \max_i \|W(i,:)\|_2. \tag{3}$$

In model reduction this quantity is the error constant of the discrete empirical interpolation method (DEIM) and controls its stability under noise [13, 14, 16]; the survey [17] points out that the spectral norm of the interpolation matrix is more informative than entrywise bounds. Strong RRQR bounds the entries of W by a constant and hence $A(S) \leq \sqrt{1 + f^2 k(n-k)}$ [6, 1], but this requires the expensive factorization and ignores the approximation error budget of an existing skeleton.

Exchange methods. Given a skeleton, one can improve it by one-for-one exchanges. Strong RRQR, maxvol and the Dominant algorithm exchange rows to increase the volume of the selected submatrix [6, 7, 8, 9]; Cortinovis and Ying combine uniform random selection with a strong-RRQR refinement [11]; Grigori and Xue sketch before a deterministic strong RRQR [12]; Kozyrev and Osinsky study subset selection that minimizes the spectral norm of a pseudoinverse [10]. These methods target the volume or a bound on W; none of them takes the approximation error of the initial skeleton as an explicit budget.

Contributions.

1. *Diagnosis.* The approximation error does not control the amplification, even for exact-rank matrices (Example 1), and for RBRP skeletons of the same size and similar error the amplification can differ by factors of 3.7 to 142 (Section 3.3), and in our experiments it can be reduced by factors between 1.25 and 3.9 (median) by cheap post-processing (Section 5).

2. *Error-budgeted refinement.* We state a deterministic, parameter-free exchange refinement in which the exact amplification is the acceptance criterion and the error budget $E \leq (1 + \varepsilon)E(S_0)$ is a hard constraint (Algorithm 1); it terminates and its full scan ends at a single-exchange local optimum (Section 4).

3. *Comparison and pipeline.* Volume-based exchange is a strong and cheap baseline but violates a 5% budget in 32% of our runs (93 of 290; 95% interval 27–38%), by up to a factor 2.4. A two-stage pipeline—guarded volume exchange followed by the guarded refinement—lowers the amplification further at comparable cost (Algorithm 2).

4. *Cost–benefit characterization.* We measure the effect of the scan budget T, of the ordering, of the rank and of the matrix size, and the effect of the error budget (Section 5).

We do not claim a new exchange rule: the ordering used below coincides with the selection rule of [9] for orthonormal bases, and we do not claim an accuracy guarantee comparable to those of strong RRQR.

Outline. Section 2 reviews related work. Section 3 introduces the setting, a conceptual example, diagnostic evidence for RBRP skeletons and the algorithms. Section 4 collects the properties we can prove. Section 5 reports the experiments and Section 6 discusses limitations.

2 Related work

Interpolative decompositions and CUR. The interpolative decomposition (ID) with bounded interpolation coefficients was introduced in [1]; CUR and skeleton decompositions are treated in [34, 35, 15]. The numerical stability of CUR approximations, including the effect of oversampling, is analysed in [25], and CUR has been used for parameter-dependent matrices [26]. IDs are used in applications such as multimodal data [27]; see [28, 24] for surveys of randomized low-rank methods and their scientific applications. Blind error estimation for CUR approximations is studied in [29]; we evaluate E and A exactly.

Randomized pivoting. Randomly pivoted Cholesky [19] and its accelerated version [20], randomized pivoting for CUR and ID [3], RBRP [2] and its partially pivoted LU variants [21], and randomized QR with column pivoting [4] select skeletons adaptively from residuals or sketches at low cost. Adaptive randomized pivoting [22] is a randomization of Osinsky's deterministic algorithm for column subset selection and is applied to DEIM, cross approximation and Nyström approximation; [23] relates it to volume sampling and gives faster implementations. The guarantees and experiments in this line concern the approximation error and the skeleton size. To our knowledge none of these works studies the interpolation matrix of the returned skeleton as an optimization target, or post-processing of a given skeleton under a budget on the approximation error.

Strong rank-revealing QR. Strong RRQR [6] computes a column (or row) subset whose interpolation coefficients are bounded entrywise, hence $\|W\|_2 \le \sqrt{1 + f^2 k(n-k)}$, by a sequence of exchanges that increase a determinant; it is the standard route to stable IDs [1]. Randomized variants combine sketching or sampling with a strong RRQR: Cortinovis and Ying [11] select rows uniformly at random, apply a strong RRQR to them, and refine the selection by alternating between rows and columns, with error bounds for perturbations of incoherent low-rank matrices; Grigori and Xue [12] apply a deterministic strong RRQR to a sketch. Damle [30] shows that Bischof–Stewart pivoting applied to a matrix with orthonormal rows yields a strong RRQR and gives a randomized variant, which is the same orthonormal-basis structure that the volume baseline of Section 3 exploits. These methods construct a skeleton; they do not take the approximation error of a given skeleton as a budget.

Volume maximization and subset selection. Maxvol [7], its rectangular version [8], and the Dominant algorithm [31] exchange columns to obtain a (locally) maximal volume, which bounds $\|X_S^\dagger X\|_F$ and $\|X_S^\dagger X\|_2$ for an exact-rank or orthonormal-basis model. Osinsky and Kozyrev [9] accelerate the exchange by splitting it: the entering column maximizes $l_j = \|(X_S^\dagger X)_j\|_2^2$ and the leaving column is chosen by volume, starting from a column-pivoted QR; Kozyrev and Osinsky [10] study the minimization of the spectral norm of the pseudoinverse of the selected submatrix, and [32] volume sampling for subset selection; see also [33]. Our ordering of the candidate rows by $\|W(i,:)\|_2$ coincides with the entering rule of [9] for an orthonormal basis; the difference is the leaving rule, which here is the exact amplification under an error budget instead of the volume, and the setting, in which the skeleton comes from RBRP and the matrix is not of exact rank k.

Interpolation stability in model reduction. In DEIM the error constant $\|(P^T U)^{-1}\|_2$ plays the role of $A(S)$ (Proposition 2); DEIM itself selects its points greedily [13], the constant is bounded a priori for points chosen by QR with column pivoting [14], and the stability of empirical interpolation under noise, with randomized and deterministic sampling points, is studied in [16]; the survey

[17] notes that the spectral norm of the interpolation matrix is more informative than entrywise bounds.

Exchange algorithms in optimal design. One-for-one exchange goes back to the exchange algorithms for D-optimal designs [36]; local search for optimal design admits approximation guarantees for some criteria but can end in arbitrarily poor local optima for others [18], which is why we state local optimality only with respect to single exchanges.

Position of this work. Relative to the above, we do not propose a new exchange rule or a new stability measure. We study the post-processing of RBRP skeletons under an explicit error budget, quantify what volume-based exchange does to the approximation error, and evaluate a two-stage pipeline that combines the cheap volume stage with an exact, spectral-norm-driven stage.

3 Algorithms

3.1 Setting

Throughout, S is *valid* if X_S has full row rank k (numerically: smallest singular value above $10^{-12}\max(\sigma_{\max}, 1)$); invalid skeletons are rejected, never regularized. The rows of W indexed by S are unit vectors, so $A(S) \geq 1$. Let S_0 be the skeleton returned by RBRP, $E_0 = E(S_0)$ and $A_0 = A(S_0)$. We study

$$\min_{|S|=k} A(S) \quad \text{subject to} \quad E(S) \leq (1+\varepsilon)E_0, \qquad \varepsilon = 0.05, \tag{4}$$

solved heuristically by one-for-one exchanges starting from S_0. We report $\rho_A = A(S_f)/A_0$ and $\rho_E = E(S_f)/E_0$.

3.2 A conceptual example

Example 1. *Let* $\varepsilon \in (0, 1]$ *and* $X = \begin{bmatrix} 1 & 0 \\ 1 & \varepsilon \\ 0 & 1 \end{bmatrix}$, *so* $k = 2$ *and* $\operatorname{rank} X = 2$. *The skeletons* $S_1 = \{1, 3\}$ *and* $S_2 = \{1, 2\}$ *both span the row space, hence* $E(S_1) = E(S_2) = 0$. *For* S_1 *we have* $X_{S_1} = I$ *and* $W = X$, *so* $A(S_1) \leq \|X\|_F = \sqrt{3+\varepsilon^2} \leq 2$. *For* S_2 *the third row is* $-\varepsilon^{-1}X(1,:) + \varepsilon^{-1}X(2,:)$, *so* $\|W(3,:)\|_2 = \sqrt{2}/\varepsilon$ *and* $A(S_2) \geq \sqrt{2}/\varepsilon$. *The two skeletons are indistinguishable by the approximation error while their amplifications differ by a factor of at least* $1/(\sqrt{2}\,\varepsilon)$, *which is unbounded as* $\varepsilon \to 0$.

3.3 Decoupling of error and amplification for RBRP skeletons

Example 1 is an exact-rank statement. We ran three diagnostic experiments, whose protocol was fixed before looking at the results, to see whether the same decoupling occurs for unmodified RBRP output (RBRP runs use block size $b = 4$). The summary statistic is the *matched gap*: among skeletons of the same size whose relative errors lie within a window of relative width w (and containing at least m skeletons), the largest ratio of the amplifications; for the RBRP experiments the maximum is taken over all ranks.

Table 1: Matched gaps of RBRP skeletons ($b = 4$, 40 seeds, one fixed matrix per family; $w = 10\%$, $m = 5$). The last column gives the range of the amplification over all recorded skeletons of the family.

Family	k	skeletons in window	error window	A in window	gap	A overall
Collinear clusters	4	9	$[0.693, 0.756]$	$[1.9, 157]$	83.7	$[1.1, 697]$
Localized SVD	12	7	$[0.0239, 0.0261]$	$[5.3, 29.5]$	5.59	$[2.5, 47.3]$
Repeated directions	6	14	$[0.324, 0.352]$	$[6.3, 895]$	142	$[3.4, 895]$
Kernel snapshots	16	8	$[0.0293, 0.0322]$	$[7.9, 29.3]$	3.70	$[4.6, 31.6]$

Exact-rank numerical example. Take $n = 200$ rows in $\mathbb{R}^2$: $(1, 0)$, $(0, 1)$, $(1, \varepsilon)$, $(1, 2\varepsilon)$ with $\varepsilon = 10^{-4}$ and 196 further rows in the same plane. The skeletons $\{(1, 0), (0, 1)\}$ and $\{(1, 0), (1, \varepsilon)\}$ both reproduce X to rounding error ($E = 0$ and 2.8×10^{-13}), but their amplifications are 14.9 and 1.23×10^5, a ratio of 8.25×10^3.

Noisy fixed-rank skeletons. For an approximately rank-2 matrix (160×20, many nearly collinear rows, noise level 2×10^{-3}) we evaluated 9000 skeletons of size two (a random subset of the $\binom{160}{2} = 12{,}720$ pairs). With $w = 5\%$ and $m = 20$, the window containing the largest gap has 1635 skeletons with errors in $[0.402, 0.423]$ and amplifications in $[6.4, 758.5]$: a ratio of 117.6.

RBRP skeletons. For each of four matrix families (the synthetic datasets of Section 5, one fixed matrix per family), RBRP was run with 40 random seeds and every accepted rank was recorded. With $w = 10\%$ and $m = 5$ we obtain the gaps of Table 1 (Figure 1). With block size $b = 8$ the same protocol gives 78.1, 6.25, 3.04 and 10.6 for collinear clusters, localized SVD, repeated directions and kernel snapshots, so the order of magnitude of the effect depends on the configuration, while a gap of at least a factor 3 is present in all eight cases. Two caveats apply. The gap is a *largest* ratio over windows that contain only 5–14 skeletons, so it shows what can happen for skeletons of equal size and similar error, not what is typical; the scatter plots in Figure 1 show the typical spread. And the diagnostic matrices are the synthetic families, not application matrices. This supports only the claim that the approximation error does not by itself control the amplification of RBRP skeletons; it is not a theorem.

3.4 Error-budgeted exchange refinement

Algorithm 1 orders the non-skeleton rows by $\|W(i,:)\|_2$ because $A(S) \geq \max_i \|W(i,:)\|_2$ by (3): the row with the largest norm is a direct source of amplification. The ordering only decides which row is tried first.

Here $\tau = 10^{-12}$ prevents floating-point cycling. With $T = \infty$ the algorithm is parameter-free and deterministic; $T < \infty$ is the budgeted variant.

3.5 Volume-based exchange and the two-stage pipeline

Let $U \in \mathbb{R}^{n \times k}$ contain the top-k left singular vectors of X and $\hat{A} = UU_S^{-1}$. Exchanging row i for s_j multiplies $|\det U_S|$ by $|\hat{A}_{ij}|$ and updates $\hat{A}$ by a rank-one formula [7]. *Vol*(f) repeatedly performs the exchange with the largest $|\hat{A}_{ij}|$ while it exceeds f ($f = 1$: locally maximal volume). It uses only U and does not look at E; afterwards we evaluate the exact A and E of the result. *VolGuard* performs the same exchanges in decreasing order of $|\hat{A}_{ij}|$ (at most 500 candidates per iteration) but accepts one only if the exact error satisfies $E(S') \leq (1 + \varepsilon)E_0$.

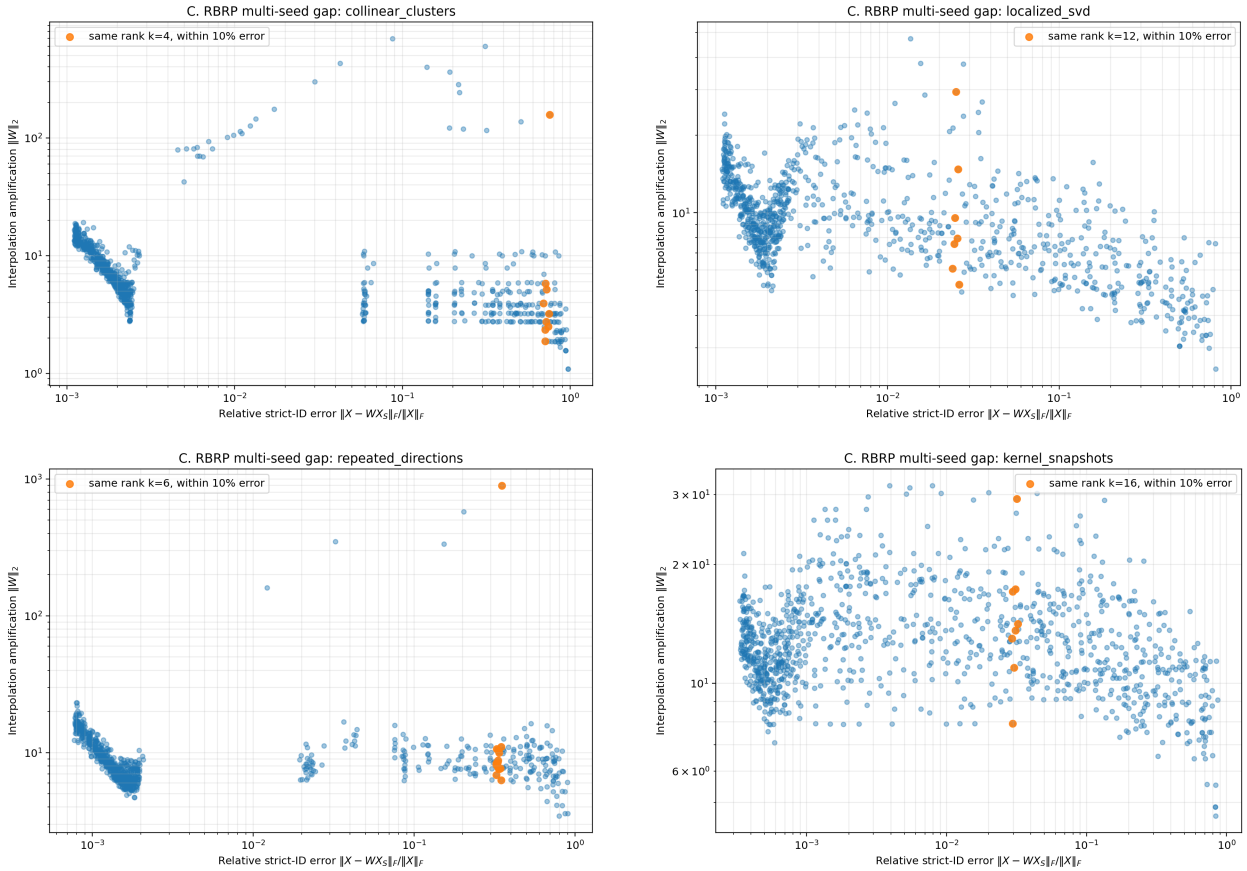


Figure 1: Error versus amplification of the RBRP skeletons of 40 seeds (all ranks); the orange points form the window of the largest matched gap.

4 Properties

Proposition 1 (Guarantees of Algorithm 1)**.** *Let S_0 be valid. (a) Every iterate is valid and satisfies $E(S) \le (1+\varepsilon)E_0$. (b) A decreases by at least a factor $(1-\tau)$ at each accepted exchange; hence the algorithm terminates after finitely many steps. (c) If $T = \infty$, the output S_f is a single-exchange local optimum: no exchange S' with $E(S') \le (1+\varepsilon)E_0$ has $A(S') < (1-\tau)A(S_f)$. (d) For $T < T'$ and the same tie-breaking, the trajectory with budget T is a prefix of that with T'; hence $A(S_f^{(T')}) \le A(S_f^{(T)})$.*

Proof. (a), (b): by construction an exchange is accepted only if it is valid, satisfies the budget and decreases A; there are finitely many k-subsets, and A cannot return to a previous value. (c): the algorithm stops only after a pass in which every non-skeleton row has been tried without an accepted exchange. (d): the two runs choose the same exchange as long as the budgeted run finds an admissible exchange among its first T rows, because both pick the first row in the order that admits one and the best exchange for that row; when the budgeted run stops, the other run continues from the same skeleton and can only decrease A further. □

Proposition 2 (Exact-rank case)**.** *If* $\operatorname{rank} X = k$ *and $U \in \mathbb{R}^{n\times k}$ has orthonormal columns spanning the column space of X, then for every valid S, $E(S) = 0$, $W = UU_S^{-1}$ and $A(S) = \|U_S^{-1}\|_2 = 1/\sigma_{\min}(U_S)$.*

Algorithm 1 Error-budgeted exchange refinement $\textsc{Refine}(X, S_0, \varepsilon, T)$

$E_0 \leftarrow E(S_0)$; $S \leftarrow S_0$
repeat
  $W \leftarrow XX_S^{\dagger}$; order $N = \{1, \dots, n\} \setminus S$ by decreasing $\|W(i,:)\|_2$ (stable ties); keep the first T entries
  for i in this order **do**
    $\mathcal{C} \leftarrow \{S' = (S \setminus \{s_j\}) \cup \{i\} : \ S' \text{ valid}, \ E(S') \le (1+\varepsilon)E_0, \ A(S') < (1-\tau)A(S), \ j = 1, \dots, k\}$
    if $\mathcal{C} \neq \emptyset$ **then** $S \leftarrow \arg\min_{S' \in \mathcal{C}} A(S')$ (ties: smaller E); **break**
    end if
  end for
until no accepted exchange in the last pass
return S

Algorithm 2 Two-stage pipeline $\textsc{Hybrid}(X, S_0, \varepsilon, T)$

$S_1 \leftarrow \text{VolGuard}(X, S_0, f{=}1, \varepsilon)$ ▷ cheap, guarded by $E_0 = E(S_0)$
$S_f \leftarrow \textsc{Refine}(X, S_1, \varepsilon, T)$ with the budget still relative to E_0
return S_f

Proof. Write $X = UB$ with $B \in \mathbb{R}^{k \times d}$ of full row rank. Then $X_S = U_S B$ with U_S invertible, $X_S^{\dagger} = B^{\dagger} U_S^{-1}$ and $W = UBB^{\dagger}U_S^{-1} = UU_S^{-1}$. Since U has orthonormal columns, $\|W\|_2 = \|U_S^{-1}\|_2$. □

Hence in the exact-rank case $A(S)$ coincides with the DEIM error constant $\|(P^T U)^{-1}\|_2$ [13, 14], and $\|W\|_2^{-1} = \sigma_{\min}(U_S)$, which is the quantity that volume maximization controls only indirectly through $\det U_S$.

Proposition 3 (Pipeline)**.** *Algorithm 2 returns a valid skeleton with $E(S_f) \le (1+\varepsilon)E_0$ and $A(S_f) \le A(S_1)$.*

This follows from Proposition 1(a),(b), because the refinement is started from S_1 with the same budget. It says that the second stage cannot hurt; the size of the gain is an empirical question (Section 5).

Proposition 4 (Cost)**.** *One exact evaluation of $(A(S'), E(S'))$ costs $O(ndk)$ operations. A pass of Algorithm 1 with budget T costs at most Tk evaluations; with $T = \infty$ the last (certifying) pass costs $(n-k)k$ evaluations, i.e. $O(n^2 dk^2)$ operations.*

Proposition 5 (Perturbed skeleton)**.** *Let $R = WX_S - X$ and let the skeleton rows be observed as $X_S + N$ with $N \in \mathbb{R}^{k \times d}$, $\hat{X} = W(X_S + N)$. (a) If the entries of N are i.i.d. $\mathcal{N}(0, \sigma^2)$, then $\mathbb{E}\|\hat{X} - X\|_F^2 = \|R\|_F^2 + \sigma^2 d \|W\|_F^2$. (b) For every $\delta > 0$ there is N with $\|N\|_F = \delta$ and $\|\hat{X} - X\|_F^2 \ge \|R\|_F^2 + \delta^2 \|W\|_2^2$, and for all such N, $\|\hat{X} - X\|_F \le \|R\|_F + \delta \|W\|_2$.*

Proof. (a) The cross term $2\langle R, WN \rangle$ has mean zero and $\mathbb{E}\|WN\|_F^2 = \sigma^2 d \|W\|_F^2$. (b) Let $W = \sum_i \sigma_i u_i v_i^T$ and $N = \delta v_1 w^T$ with $w \in \mathbb{R}^d$ a unit vector chosen so that $\langle R, u_1 w^T \rangle \ge 0$ (e.g. $w = \pm R^T u_1 / \|R^T u_1\|$). Then $\|R + WN\|_F^2 = \|R\|_F^2 + 2\delta\sigma_1 \langle R, u_1 w^T \rangle + \delta^2 \sigma_1^2$. The upper bound is the triangle inequality. □

With $\sigma = \nu \|X\|_F / \sqrt{nd}$, (a) gives the relative error $\sqrt{E^2 + \nu^2 \|W\|_F^2 / n}$. Thus the amplification $\|W\|_2$ controls the worst case, whereas random noise is controlled by $\|W\|_F \in [\sqrt{k}, \sqrt{k}\|W\|_2]$.

Table 2: Main experiment on the four synthetic datasets ($n = 500$, $d = 60$, 40 RBRP seeds each). $\rho_A = A_f/A_0$ and $\rho_E = E_f/E_0$ are averaged over seeds.

Dataset	full ρ_A	T=10 ρ_A	full ρ_E	evals (full)	evals (T=10)	time (s) full	time (s) T=10
Collinear clusters ($k = 4$)	0.522	0.573	1.007	3,215	51	2.4	0.04
Repeated directions ($k = 4$)	0.433	0.464	0.994	2,291	71	1.8	0.06
Localized SVD ($k = 14$)	0.380	0.392	0.793	7,119	296	7.8	0.32
Kernel snapshots ($k = 16$)	0.356	0.419	0.942	15,222	658	17.2	0.76
All (160 runs)	0.423	0.462	0.934	6,962	269	7.3	0.29

Remark 1. *We do not prove an accuracy guarantee for the refined skeleton. Local search can end in poor local optima in general (cf. the analysis of exchange algorithms for optimal designs in [18]), and Proposition 1(c) only says that no single exchange within the budget helps.*

5 Numerical experiments

5.1 Setup

Matrices. Four synthetic datasets with $n = 500$, $d = 60$ (nearly collinear clusters, repeated directions, localized SVD, kernel snapshots) at representative ranks $k = 4, 4, 14, 16$; a Burgers snapshot matrix (1000×256: 25 initial phases $\times$ 8 viscosities $\times$ 5 times of the periodic viscous Burgers equation, pseudo-spectral with a first-order IMEX step; $k = 8, 16, 24$); and the scikit-learn handwritten digits (1797×64; $k = 10, 20, 30$).

Protocol. All methods start from the same RBRP skeleton ($b = 4$); $\varepsilon = 0.05$; A and E are always evaluated exactly on the full matrix. Seeds: 40 per synthetic dataset in Section 5.2, 20 per configuration elsewhere; full scans on Burgers and digits k=10 use 10 seeds. QRCP denotes QR with column pivoting applied to X^T. Intervals are 95% percentile bootstrap intervals (5000 resamples of the seeds within each configuration, keeping the pairing of the methods); pooled statistics are recomputed for every resample. The intervals quantify the randomness of RBRP for the given matrices; they do not capture the variation across matrices, which is only represented by the 19 configurations used. Timings are single-threaded wall-clock times on one machine.

5.2 Main experiment

Table 2 shows that the full scan lowers the mean amplification to 42% of the RBRP value with the error budget satisfied in all runs, and that the budgeted variant ($T = 10$) obtains most of this at a small fraction of the evaluations. In all 160 runs the full scan ended at a single-exchange local optimum, and an independent brute-force check of all $k(n - k)$ exchanges on 8 runs found no admissible exchange.

5.3 Scan budget and ordering

Table 3 and Figure 2: ρ_A decreases monotonically with T (by Proposition 1(d) this holds run by run), with strongly diminishing returns. At matched cost, ordering by $\|W(i,:)\|_2$ is far better than a random order (for $T = 10$: 0.462 versus 0.809 with 269 versus 257 evaluations); the row-norm order was better in 89–96% of the paired runs. $T = 10$ retains about 93% of the full-scan reduction at

Table 3: Scan budget T and ordering ablation (160 runs, same S_0 for both orders).

T	row-norm order ρ_A (SE)	evals	random order ρ_A (SE)	evals
1	0.575 (0.020)	33	0.948 (0.011)	13
2	0.535 (0.019)	57	0.923 (0.012)	30
5	0.483 (0.017)	143	0.857 (0.015)	88
10	0.462 (0.017)	269	0.809 (0.016)	257
20	0.448 (0.017)	591	0.728 (0.017)	581
50	0.438 (0.017)	1,278	0.618 (0.018)	1,818
∞	0.423 (0.016)	6,962	–	–

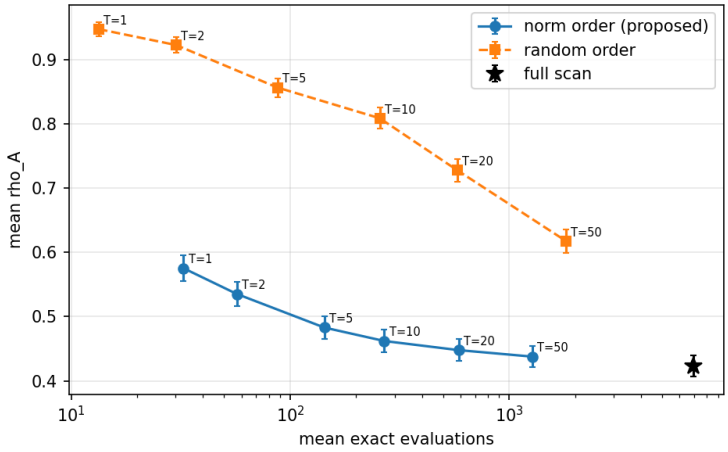


Figure 2: Cost–benefit of the scan budget T: mean ρ_A versus mean number of exact evaluations, for the row-norm order and for a random order.

4% of its evaluations; this value was chosen from the same sweep and is a representative operating point, not a tuned constant.

5.4 Rank sensitivity

Table 4 and Figure 3: the reduction is present at every tested rank (mean $\rho_A < 1$ in all 16 configurations); there is no systematic trend with k, and differences between neighbouring ranks are mostly within one or two standard errors.

5.5 Comparison with volume-based exchange and the pipeline

Tables 5 and 6 and Figure 4 give the main comparison.

- Every post-processing method lowers the median amplification of the RBRP skeleton; the median over all runs of $A(\text{RBRP})/A(\text{pipeline})$ is 1.78 (95% interval 1.70–1.90).
- Unguarded volume exchange violates the budget in 64 of 200 runs (32%, 95% Wilson interval 26–39%; maximum $E = 2.44E_0$; all runs on Burgers $k = 8, 16$); the guarded methods satisfy it in all runs.
- At $T = 10$ the refinement alone is not better than guarded volume exchange: the median A ratio is 0.997 (0.985–1.004, no detectable difference) and the median error ratio is 1.067

Table 4: Rank sensitivity (20 seeds per configuration).

Dataset	k	median A_0	full ρ_A (SE)	T=10 ρ_A (SE)
Kernel snapshots	8	12.3	0.475 (0.033)	0.509 (0.038)
Kernel snapshots	12	14.4	0.401 (0.029)	0.447 (0.041)
Kernel snapshots	16	15.1	0.348 (0.033)	0.401 (0.042)
Kernel snapshots	20	15.4	0.391 (0.031)	0.440 (0.036)
Localized SVD	6	15.9	0.333 (0.042)	0.347 (0.045)
Localized SVD	10	17.0	0.347 (0.047)	0.355 (0.046)
Localized SVD	14	17.3	0.313 (0.037)	0.325 (0.039)
Localized SVD	18	15.6	0.374 (0.038)	0.393 (0.041)
Collinear clusters	4	5.1	0.498 (0.048)	0.556 (0.057)
Collinear clusters	6	4.4	0.520 (0.049)	0.556 (0.052)
Collinear clusters	8	5.6	0.474 (0.038)	0.485 (0.038)
Collinear clusters	12	5.1	0.576 (0.051)	0.576 (0.051)
Repeated directions	4	9.8	0.446 (0.021)	0.474 (0.020)
Repeated directions	6	11.4	0.433 (0.032)	0.450 (0.033)
Repeated directions	8	10.7	0.490 (0.026)	0.490 (0.026)
Repeated directions	10	7.3	0.530 (0.037)	0.530 (0.037)

Table 5: Median amplification A after each method (20 RBRP seeds; the full scan uses 10 seeds on Burgers and digits k=10 and none on digits k=20, 30). The last column is the share of Vol(1) runs satisfying $E \leq 1.05E_0$.

Matrix	k	RBRP	Vol(1)	VolGuard	Ours-T10	Hybrid	Ours-full	Vol(1) feasible
Burgers	8	17.7	13.2	13.1	13.1	12.4	9.3	0%
Burgers	16	21.4	8.0	12.8	13.6	11.1	8.1	0%
Burgers	24	26.9	11.6	12.3	12.6	12.2	10.8	35%
Digits	10	17.6	15.2	15.2	14.8	14.0	12.5	100%
Digits	20	18.9	15.7	15.7	15.1	14.0	–	100%
Digits	30	24.3	16.0	16.0	16.8	14.4	–	100%
Kernel snapshots	16	15.1	6.4	6.1	6.3	5.9	5.6	55%
Localized SVD	14	17.3	4.8	4.8	4.7	4.4	4.5	95%
Collinear clusters	4	5.1	2.6	2.6	2.4	2.6	2.2	100%
Repeated directions	4	9.8	4.5	4.5	4.5	4.3	4.2	95%

(1.050–1.078); the guarded volume exchange dominates in 30% (24–36%) of the pairs, the refinement in 7.5% (4–11%).

- The pipeline improves on the guarded volume exchange: median A ratio 0.963 (0.937–0.976) at an unchanged error (ratio 1.001, 1.000–1.007); it dominates in 18% (13.5–23%) of the pairs and, by construction, is never dominated. Against the refinement alone the median A ratio is 0.946 (0.927–0.967); the pipeline dominates in 40.5% (34.5–46.5%) of the pairs and is dominated in 5.5% (2.5–9%). The gain is concentrated: it is largest on Burgers $k = 16$ (median ratio 0.857; interval 0.81–0.995) and about 8–10% on digits, while the pipeline coincides with the guarded volume exchange in 40–60% of the runs on Burgers $k = 24$, collinear clusters and repeated directions. The order matters: on collinear clusters the refinement started from the RBRP skeleton is better than the pipeline, because the volume stage ends in a local volume maximum from which the refinement finds nothing.

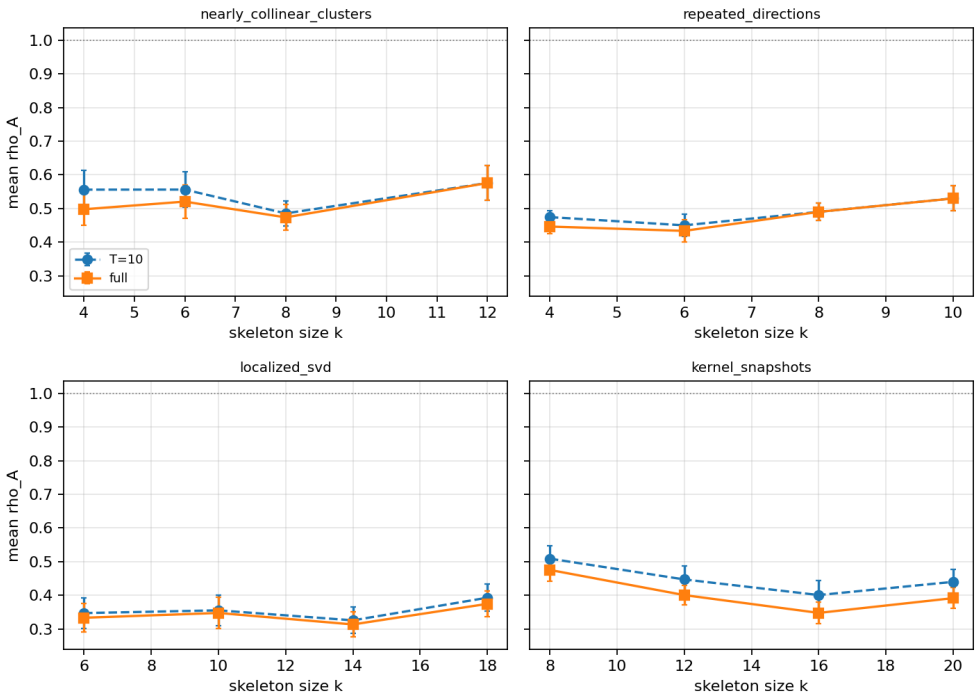


Figure 3: Mean ρ_A versus the skeleton size k ($T = 10$ and full scan).

- A full scan is clearly better under the same budget: median A ratio 0.874 (0.843–0.896) against the guarded volume exchange and 0.921 (0.894–0.945) against the pipeline, at roughly 30–240 times the cost of the pipeline.

As the sign of the difference between the pipeline (or the full scan) and the method it starts from is determined by construction, we report intervals for the size of the gain rather than significance tests.

5.6 Error budget

Table 7 and Figure 5: on the Burgers matrix the budget is a genuine trade-off: removing it lowers the median amplification by a further 24–45% but raises the median error by 39–159% (up to $5.8E_0$); a moderate budget $g = 1.20$ already gives much of the gain. On digits (with $T = 10$) the budget has almost no effect. $\varepsilon = 0.05$ is a working point, not an optimum.

5.7 Cost and scaling

The refinement is substantially more expensive than RBRP itself: with $T = 10$ it takes 17–281 times the RBRP time for $n \leq 4000$ and 0.01–1.8 s; the full scan ($n \leq 2000$) takes 800–17000 times RBRP. Fitted exponents of time in n: RBRP 0.72–0.86, $T = 10$ 0.94–1.15 (the number of evaluations does not grow with n while each evaluation is linear in n), full scan 1.81–2.04, in agreement with Proposition 4; the exponents rest on three or four values of n and few seeds and should be read as orders of magnitude.

Table 6: Paired comparison pooled over all configurations and seeds, with 95% percentile bootstrap intervals (seeds resampled within configuration). "x dominates y" means no worse in both A and E and strictly better in one.

x	y	pairs	median A ratio	median E ratio	x dominates (%)	y dominates (%)
Ours-T10	Vol(f=1)	200	0.998 [0.992, 1.013]	1.055 [1.041, 1.070]	13.0 [9.0, 17.0]	23.0 [18.0, 28.0]
Ours-T10	VolGuard-500	200	0.997 [0.985, 1.004]	1.067 [1.050, 1.078]	7.5 [4.0, 11.0]	30.0 [24.0, 36.0]
Hybrid-T10	VolGuard-500	200	0.963 [0.937, 0.976]	1.001 [1.000, 1.007]	18.0 [13.5, 23.0]	0.0 [0.0, 0.0]
Hybrid-T10	Ours-T10	200	0.946 [0.927, 0.967]	0.983 [0.970, 0.990]	40.5 [34.5, 46.5]	5.5 [2.5, 9.0]
Hybrid-T10	Vol(f=1)	200	0.973 [0.952, 0.984]	1.000 [1.000, 1.000]	25.0 [20.0, 30.0]	0.0 [0.0, 0.0]
Ours-T10-noG	Vol(f=1)	200	0.936 [0.920, 0.953]	1.095 [1.086, 1.112]	16.5 [13.0, 20.0]	24.0 [18.5, 29.5]
Ours-full	VolGuard-500	120	0.874 [0.843, 0.896]	1.060 [1.043, 1.088]	22.5 [17.5, 28.3]	7.5 [3.3, 12.5]
Ours-full	Hybrid-T10	120	0.921 [0.894, 0.945]	1.021 [1.008, 1.034]	25.0 [19.2, 30.8]	11.7 [6.7, 16.7]
Ours-full	Vol(f=1)	120	0.903 [0.882, 0.926]	1.054 [1.024, 1.085]	30.8 [25.0, 37.5]	6.7 [3.3, 10.8]

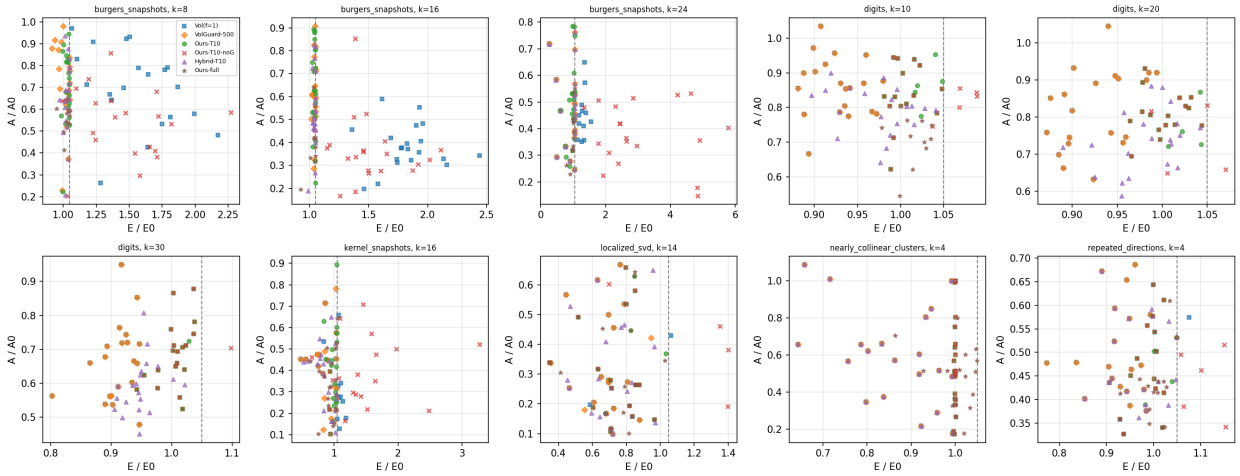


Figure 4: Per-run $(E/E_0, A/A_0)$ for the compared methods; the dashed line is the budget 1.05.

5.8 Refining a QRCP skeleton

Table 9: the refinement also lowers the amplification of a QRCP skeleton (full scan by 16–41%; $T = 10$ by 0–23%, with essentially no effect on Burgers $k = 8, 24$). This is a single deterministic run per row and is only indicative.

5.9 Downstream effect of the amplification

We evaluate the reconstruction $\hat{X} = W(X_S + N)$ with the closed forms of Proposition 5; a Monte-Carlo check with 100 draws agrees with formula (a) to within about 1% in all 20 tested cases. Over all skeletons, $\log \|W\|_2$ and $\log \|W\|_F$ are strongly correlated (correlation 0.94) and $\|W\|_2/\|W\|_F$ has median 0.55.

- *Lower amplification lowers the error under perturbations (Table 10).* At a large perturbation, the pipeline reduces the median reconstruction error of the RBRP skeleton by 27% for random noise ($\nu = 1$; interval of the ratio 0.705–0.746) and by 37% in the worst case ($\eta = 1$; 0.595–0.660); the Frobenius norm $\|W\|_F$ falls in 98.5% of the runs (median ratio 0.67, against 0.56 for $\|W\|_2$).

Table 7: Error-budget ablation (RBRP skeletons, T=10, 10 seeds). Each cell: median A / mean ρ_E.

Matrix	k	g=1.00	g=1.05	g=1.20	g=∞
Burgers	8	13.40 / 0.981	12.69 / 1.030	10.96 / 1.166	10.06 / 1.526
Burgers	16	14.70 / 0.982	13.15 / 1.039	9.71 / 1.181	6.91 / 1.514
Burgers	24	13.11 / 0.943	13.11 / 0.975	12.17 / 1.045	9.87 / 3.264
Digits	10	14.83 / 0.988	15.00 / 1.011	14.64 / 1.023	14.64 / 1.023
Digits	20	15.38 / 0.982	15.66 / 1.017	15.39 / 1.007	15.39 / 1.007
Digits	30	17.49 / 0.984	16.51 / 1.001	16.43 / 1.008	16.43 / 1.008

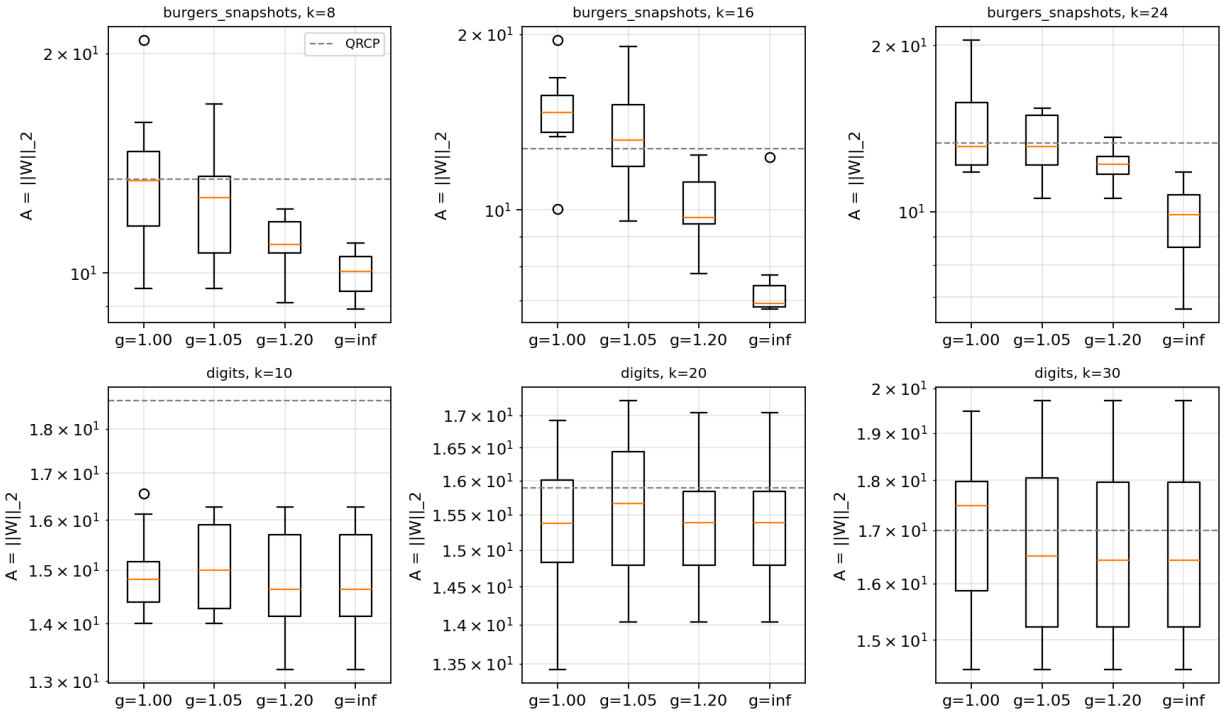


Figure 5: Amplification for different budgets g (dashed: QRCP).

- *The benefit appears only above a matrix-dependent noise level (Table 11).* The crossover ν^* ranges from about 0.004 to 2.3 (relative to the rms entry of X): for localized SVD, kernel snapshots and the Burgers matrix at $k = 24$ the benefit is visible at noise of a few percent, for the collinear and repeated-direction matrices only for noise comparable to the signal. For small perturbations the error is dominated by E, and the differences in Table 10 at 0.01 are mostly differences in E.

- *The second stage adds little to the perturbed reconstruction.* Paired against the guarded volume exchange, the error ratio of the pipeline is 1.000 with intervals inside $[0.999, 1.008]$ for all random-noise levels and for worst-case perturbations up to $\eta = 0.1$; only at $\eta = 1$ is it smaller, by 3.6% (0.948–0.974). So the additional reduction of $\|W\|_2$ obtained by the refinement pays off for the perturbed reconstruction only for large worst-case perturbations. The refinement alone at $T = 10$ is worse than the pipeline and the guarded volume exchange under random noise.

Table 8: Wall-clock time (single thread, $d = 60$, representative k).

Dataset	n	RBRP (ms)	T=10 (s)	T=10/RBRP	full (s)
Kernel snapshots	500	2.1	0.28	133	6.4
Kernel snapshots	1000	2.9	0.54	184	18.9
Kernel snapshots	2000	4.7	1.33	281	81.0
Kernel snapshots	4000	11.3	1.81	160	–
Localized SVD	500	1.8	0.11	61	2.4
Localized SVD	1000	2.5	0.18	75	7.6
Localized SVD	2000	4.2	0.51	123	41.2
Localized SVD	4000	10.9	1.10	101	–
Collinear clusters	500	0.8	0.01	17	0.6
Collinear clusters	1000	0.7	0.02	34	1.8
Collinear clusters	2000	1.4	0.05	35	8.9
Collinear clusters	4000	3.3	0.11	35	–
Repeated directions	500	0.6	0.02	31	0.5
Repeated directions	1000	0.7	0.02	31	1.3
Repeated directions	2000	1.0	0.05	50	6.4
Repeated directions	4000	3.0	0.14	48	–

Table 9: Refining the QRCP skeleton (one deterministic run per row).

Matrix	k	A (QRCP)	T=10: A/A_{QRCP}	E/E_{QRCP}	full: A/A_{QRCP}	E/E_{QRCP}
Burgers	8	13.44	1.000	0.999	0.789	1.014
Burgers	16	12.73	0.887	1.038	0.589	1.017
Burgers	24	13.30	1.000	1.000	0.837	1.024
Digits	10	18.66	0.769	1.001	0.676	1.002
Digits	20	15.90	0.907	1.048	–	–
Digits	30	17.00	0.932	1.023	–	–

- *Unguarded volume exchange can hurt for small perturbations.* On the Burgers matrix at $k = 8, 16$ its error stays above that of the RBRP skeleton until $\eta \approx 0.03$–0.1 because it increases E (Figure 8).

These are medians of ratios pooled over configurations; the proxy does not replace an application study (e.g. a full DEIM solve).

5.10 Additional real matrices

To test whether the findings carry over to public real data we added three matrices, with ranks $k = 8, 16, 32$ and 10 RBRP seeds each: *image patches*, 3000 random 12×12 gray-scale patches (values in $[0, 1]$) from four scikit-image test images, a 3000×144 matrix (optimal relative rank-k error $0.10, 0.078, 0.058$); *ECG*, the delay-embedding matrix (windows of 128 samples, stride 10, 3588 rows) of the first 36,000 standardized samples of the ECG record distributed with SciPy $(0.26, 0.14, 0.045)$; *digits kernel*, the 1000×1000 Gaussian kernel matrix (bandwidth equal to the median pairwise distance) of 1000 randomly chosen handwritten digits $(0.045, 0.023, 0.010)$. The protocol is that of the previous experiments; no full scan was run. Tables 12–14 and Figure 9 show:

- Unguarded volume exchange violates the budget in 29 of 90 runs (32%, Wilson interval 24–42%; maximum $E = 1.39E_0$), in all runs on image patches at $k = 8$ and on ECG at $k = 32$;

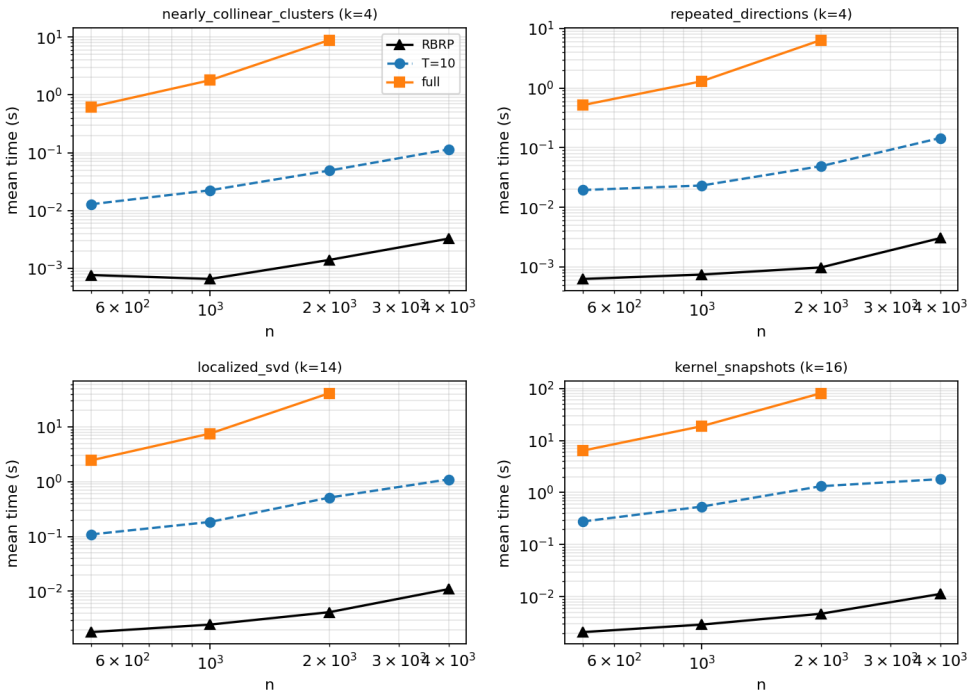


Figure 6: Mean wall-clock time versus n (log–log).

together with Section 5 this is 93 of 290 runs (32%). The guarded methods satisfy the budget in all runs.

- The refinement alone at $T = 10$ is again not better than the guarded volume exchange (median A ratio 1.030, interval 0.981–1.054; error ratio 1.099, 1.075–1.114; the guarded volume exchange dominates in 35.6% of the pairs, the refinement in 5.6%) and stalls at larger ranks: on ECG at $k = 32$ its median amplification is 20.9 against 16.0 for VolGuard and 15.0 for the pipeline.

- The pipeline gains more over the guarded volume exchange than on the earlier matrices: median A ratio 0.893 (0.866–0.922), smaller in 88% of the pairs at an error ratio of 1.029 (1.017–1.042); against the refinement alone 0.896 (0.880–0.922), dominating in 48.9% of the pairs. The gain is concentrated on image patches (ratios 0.74–0.76 at all three ranks) and on the digits kernel at $k = 32$ (0.86); on ECG at $k = 8$ the two coincide in 80% of the runs. The pooled median of ρ_A is 0.49 (0.46–0.56).

- Compared with QRCP, which is not a competitive baseline here (its amplification is 28 on image patches), the pipeline has a smaller amplification in 97% of the runs (median A/A_{QRCP} between 0.37 and 0.90); its error is between 0.72 and 1.16 times that of QRCP, 8–16% higher on image patches, and in 44% of the runs it is better in both quantities.

- The refinement is expensive on these larger matrices (up to 26 s for $T = 10$ at $n = 3000$, $k = 32$; pipeline up to 19 s), whereas Vol(1) takes at most 0.64 s.

Table 10: Reconstruction error under perturbations of the skeleton, relative to the RBRP skeleton: median over all configurations and 20 seeds of err(method)/err(RBRP), with 95% bootstrap intervals. Random noise: relative level ν; worst case: relative size η (lower bound of Proposition 5). ρ_A, ρ_F: median ratios of $\|W\|_2$, $\|W\|_F$. Last row: paired ratio of the pipeline to the guarded volume exchange. For 0.01 the differences are dominated by those in E.

	ρ_A	ρ_F	random noise ν		worst case η	
Method			0.1	1	0.1	1
Vol(1)	0.56 [0.52,0.58]	0.64 [0.60,0.68]	0.86 [0.85,0.88]	0.71 [0.70,0.74]	0.75 [0.73,0.78]	0.64 [0.59,0.68]
VolGuard	0.60 [0.57,0.64]	0.68 [0.66,0.70]	0.85 [0.84,0.86]	0.73 [0.71,0.75]	0.76 [0.73,0.78]	0.68 [0.64,0.71]
Ours-$T10$	0.61 [0.57,0.63]	0.72 [0.69,0.77]	0.96 [0.94,0.96]	0.81 [0.79,0.83]	0.81 [0.79,0.86]	0.69 [0.65,0.73]
Hybrid	0.56 [0.53,0.59]	0.67 [0.65,0.69]	0.86 [0.85,0.89]	0.73 [0.70,0.75]	0.73 [0.70,0.76]	0.63 [0.59,0.66]
Hybrid/VolGuard (paired)	–	–	1.000 [1.000,1.004]	1.000 [1.000,1.000]	1.000 [0.999,1.000]	0.964 [0.948,0.974]

Table 11: Crossover noise level $\nu^* = E_0\sqrt{n}/\|W_0\|_F$ of the RBRP skeletons (median and quartiles over 20 seeds): the random-noise term of the reconstruction error exceeds the approximation error only for $\nu > \nu^*$.

Matrix	k	median ν^*	quartiles
Burgers	8	0.134	[0.105, 0.163]
Burgers	16	0.0379	[0.0316, 0.0443]
Burgers	24	0.00366	[0.0031, 0.00461]
Digits	10	0.472	[0.451, 0.488]
Digits	20	0.252	[0.248, 0.257]
Digits	30	0.135	[0.131, 0.14]
Kernel snapshots	16	0.027	[0.0209, 0.033]
Localized SVD	14	0.0132	[0.0107, 0.0213]
Collinear clusters	4	2.27	[1.79, 3.04]
Repeated directions	4	0.949	[0.884, 1.1]

6 Discussion

We studied post-processing of RBRP skeletons under an explicit error budget. The main findings are that the amplification of RBRP skeletons is large and not controlled by the approximation error; that post-processing should respect an error budget, because unguarded volume exchange can inflate the error considerably on snapshot-type matrices; and that a two-stage pipeline gives a modest additional reduction of the amplification over guarded volume exchange (median ratio 0.963, interval 0.937–0.976; 0.893, 0.866–0.922 on three additional real matrices) at similar cost, while a full scan gives a larger reduction at a much higher cost.

Limitations. (i) The contribution is relative to RBRP: QRCP is faster than the refinement; on the matrices of Section 5 it gives an amplification comparable to that after the budgeted refinement, whereas on the three additional real matrices the pipeline has a 10–63% smaller amplification than QRCP, at an error up to 16% higher; reasons to start from RBRP (scalability, randomization) were not tested here. (ii) We prove termination and local optimality, not an accuracy guarantee, and not a bound comparable to strong RRQR. (iii) The refinement costs one to two orders of magnitude more than RBRP; cheaper evaluation (e.g. incremental updates) was not explored. (iv) The volume baseline uses the SVD basis, is not the original strong RRQR implementation, and VolGuard is our own variant. (v) The downstream evidence is a perturbation proxy: the benefit for random noise is similar to that of guarded volume exchange and appears only above a matrix-dependent noise

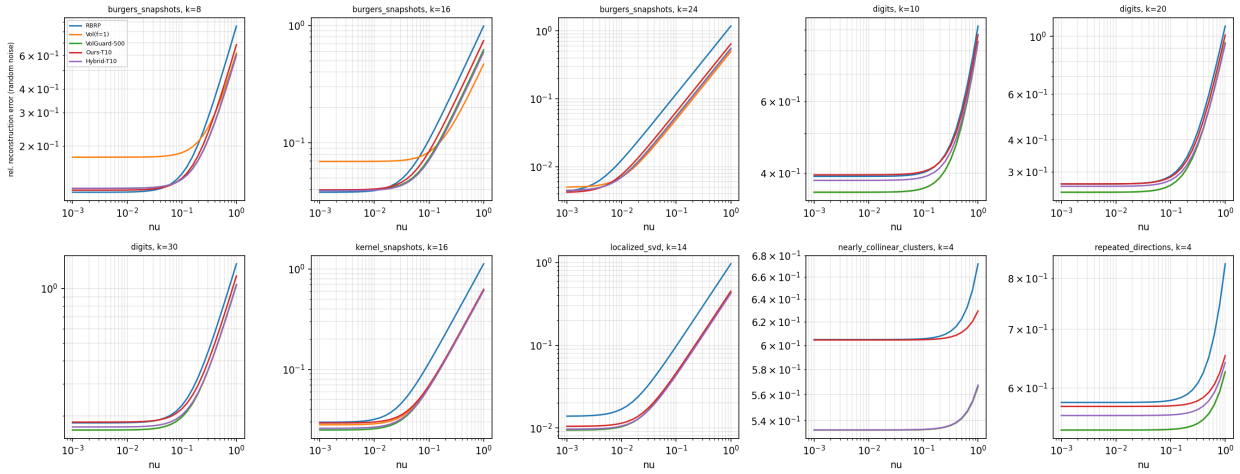


Figure 7: Expected relative reconstruction error under random noise of relative level ν (median skeleton per method).

Table 12: Additional real matrices (10 RBRP seeds): median amplification A after each method and the share of Vol(1) runs satisfying $E \leq 1.05E_0$.

Matrix	k	RBRP	QRCP	Vol(1)	VolGuard	Ours-T10	Hybrid	Vol(1) feasible
Digits kernel	8	20.9	14.2	13.0	13.0	13.2	12.7	100%
Digits kernel	16	18.2	12.3	11.9	11.9	12.0	10.9	100%
Digits kernel	32	15.4	10.6	11.2	11.2	10.7	9.6	100%
ECG Hankel	8	21.5	15.6	13.1	13.1	13.2	12.9	100%
ECG Hankel	16	26.0	17.7	14.7	14.7	14.2	13.0	100%
ECG Hankel	32	36.9	17.4	15.4	16.0	20.9	15.0	0%
Image patches	8	33.5	28.1	22.2	24.3	32.3	16.3	0%
Image patches	16	33.8	28.5	15.6	16.3	22.0	12.3	40%
Image patches	32	33.9	27.9	13.2	13.0	10.3	10.2	70%

level; (vi) Data are synthetic matrices, one snapshot matrix generated by us (whose floating-point generation makes the results reproducible only to a few percent across machines), and four small public real matrices (digits, image patches, an ECG delay matrix, a digits kernel matrix; at most 3600×1000) whose construction (patch size, bandwidth, window length) we chose; the full scan has only 10 seeds on Burgers and digits k=10 and none on digits k=20, 30; the budget $T = 10$ and $\varepsilon = 0.05$ are working points.

Future work. Combining the pipeline with other initializations, an incremental implementation of the exact evaluations, and a probabilistic analysis for incoherent low-rank-plus-noise matrices.

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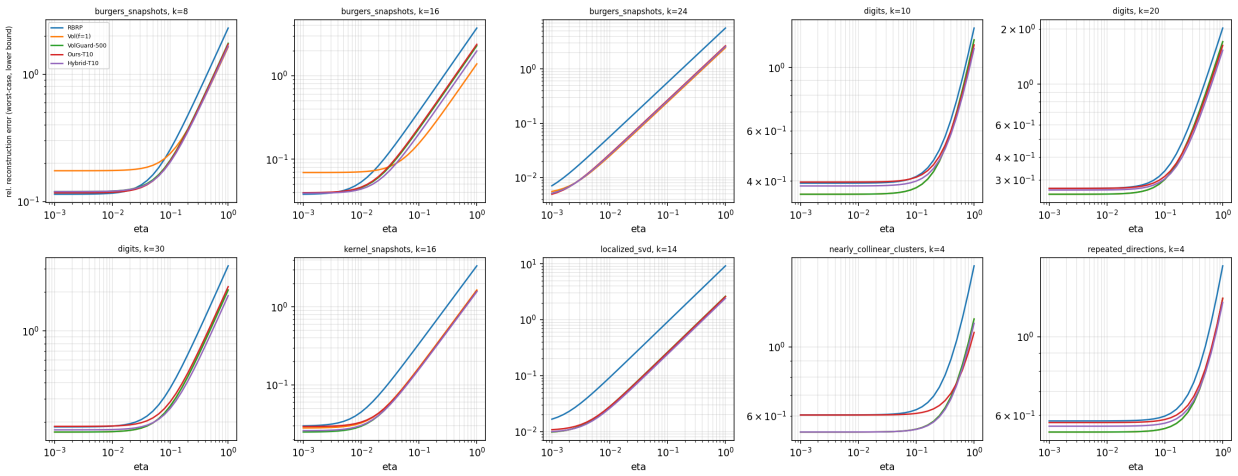


Figure 8: Lower bound on the worst-case relative reconstruction error for perturbations of relative size η.

Table 13: Pipeline versus QRCP (median over seeds of the ratios; QRCP is a single deterministic skeleton) and mean wall-clock times.

Matrix	k	Hybrid A/A_{QRCP}	Hybrid E/E_{QRCP}	Vol(1) time (s)	Ours-T10 time (s)	Hybrid time (s)
Digits kernel	8	0.89	1.02	0.63	2.3	2.2
Digits kernel	16	0.89	1.01	0.63	5.7	5.5
Digits kernel	32	0.90	1.01	0.64	25.7	19.0
ECG Hankel	8	0.83	0.90	0.09	2.2	0.9
ECG Hankel	16	0.73	0.97	0.10	7.2	3.3
ECG Hankel	32	0.86	0.72	0.14	20.3	9.6
Image patches	8	0.58	1.16	0.04	1.1	1.7
Image patches	16	0.43	1.10	0.05	5.0	4.7
Image patches	32	0.37	1.08	0.06	26.4	16.5

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Table 14: Additional real matrices: paired comparison pooled over the nine configurations and ten seeds, with 95% bootstrap intervals (as in Table 6).

x	y	pairs	median A ratio	median E ratio	x dominates (%)	y dominates (%)
Ours-T10	Vol(f=1)	90	1.031 [0.980, 1.062]	1.093 [1.061, 1.106]	7.8 [3.3, 12.2]	26.7 [20.0, 33.3]
Ours-T10	VolGuard-500	90	1.030 [0.981, 1.054]	1.099 [1.075, 1.114]	5.6 [2.2, 10.0]	35.6 [26.7, 44.4]
Hybrid-T10	VolGuard-500	90	0.893 [0.866, 0.922]	1.029 [1.017, 1.042]	16.7 [11.1, 23.3]	0.0 [0.0, 0.0]
Hybrid-T10	Ours-T10	90	0.896 [0.880, 0.922]	0.985 [0.964, 0.997]	48.9 [38.9, 57.8]	7.8 [3.3, 12.2]
Hybrid-T10	Vol(f=1)	90	0.917 [0.885, 0.943]	1.015 [1.010, 1.033]	23.3 [16.7, 30.0]	0.0 [0.0, 0.0]
Ours-T10-noG	Vol(f=1)	90	0.947 [0.924, 0.984]	1.119 [1.099, 1.126]	6.7 [2.2, 11.1]	28.9 [21.1, 36.7]

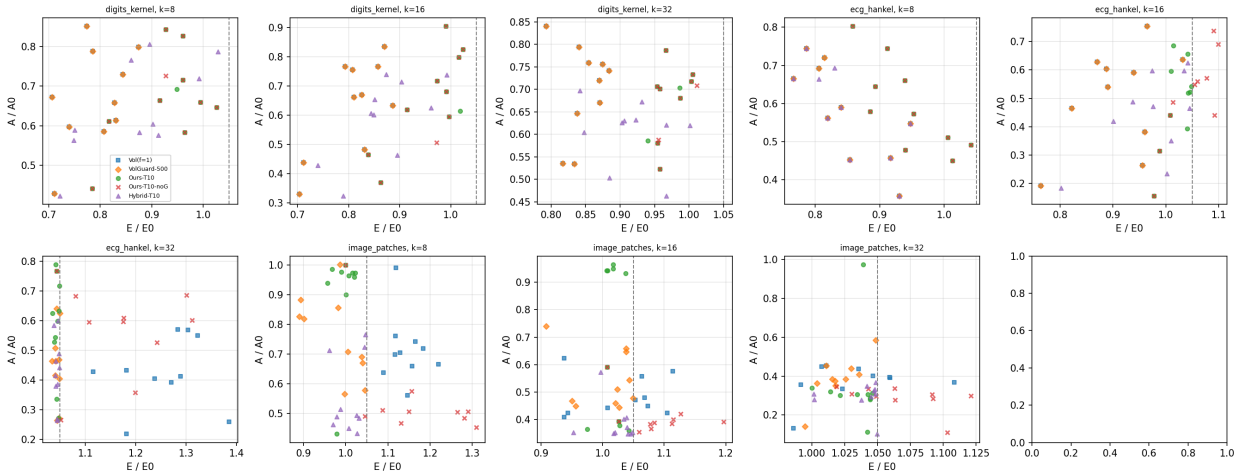


Figure 9: Per-run $(E/E_0, A/A_0)$ on the additional real matrices; the dashed line is the budget 1.05.

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